

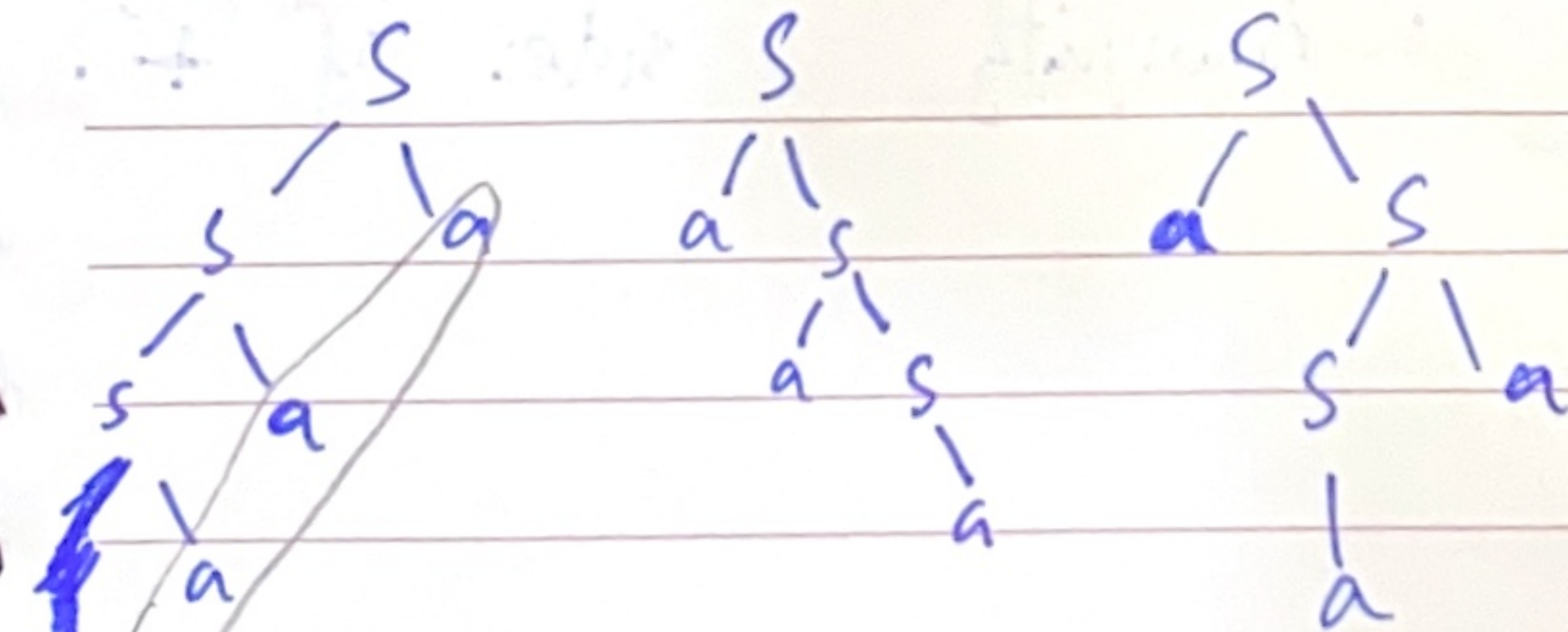
LECTURE-15

Tuesday : ONLINE
14/10/25

□ Ambiguous Grammar: One grammar produces more than one trees. Tree shouldn't grow on both sides.

① $S \rightarrow aS / Sa$

"aaa"



• Cause of : two loops for same Ambiguity terminal

• Removing : Sa / n OR (aS / n) Ambiguity

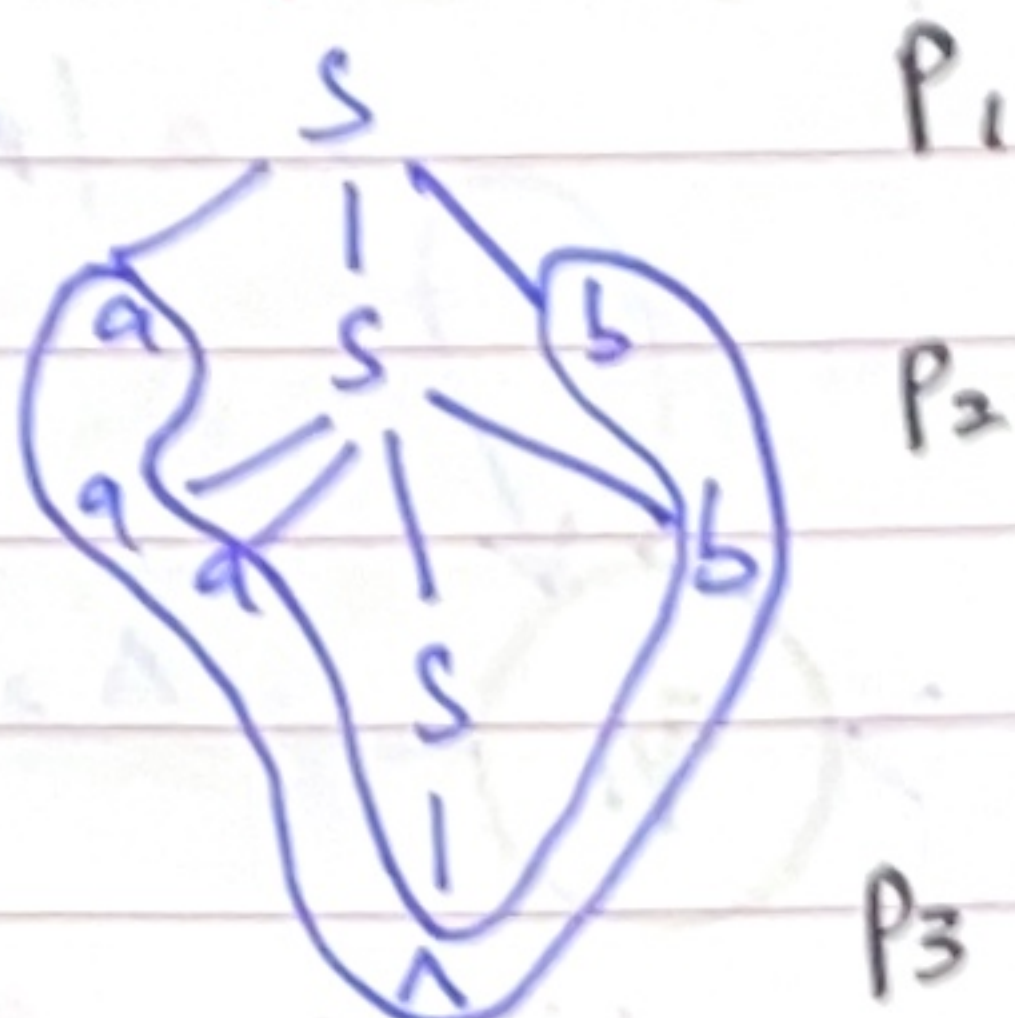
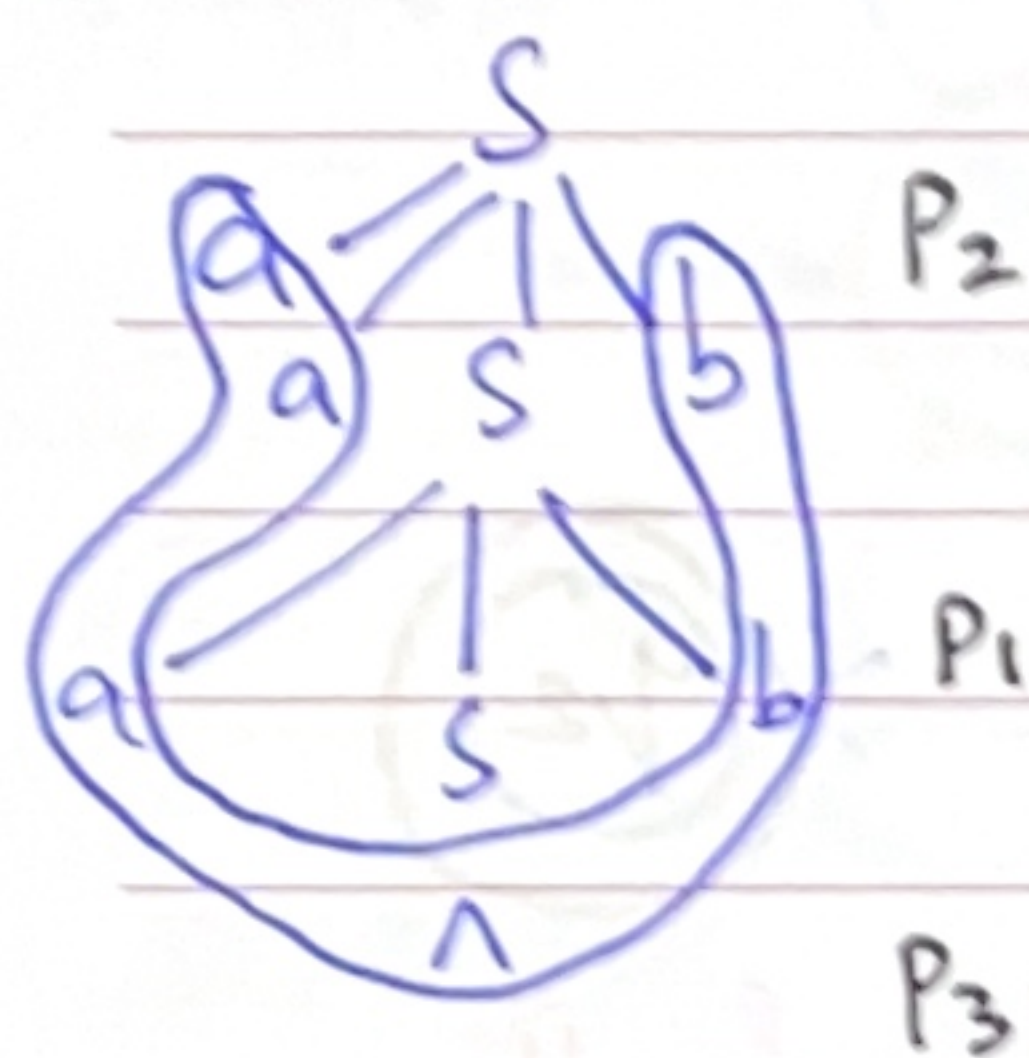


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day / date:

② $S \rightarrow a^{P_1} S^{P_2} b^{P_3} / \Lambda$

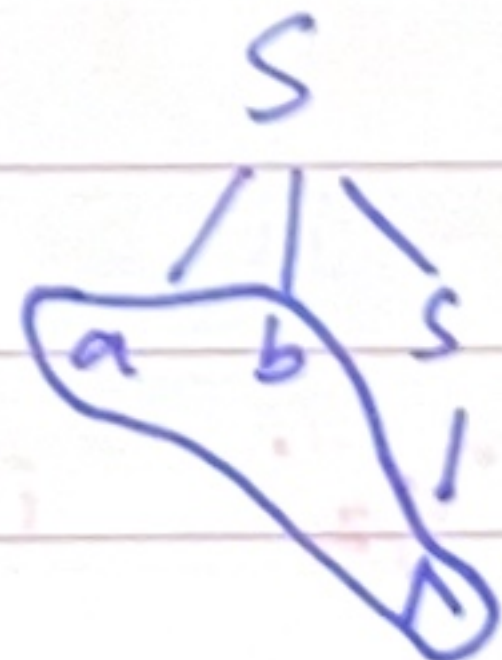
- Cause of Ambiguity:



"aaabbb"

- Removing Ambiguity : $S \rightarrow aSb / \bar{T}$
 $\bar{T} \rightarrow aaTb / \Lambda$

③ $S \rightarrow aSb / abS / \lambda$: Ambiguous on "ab".

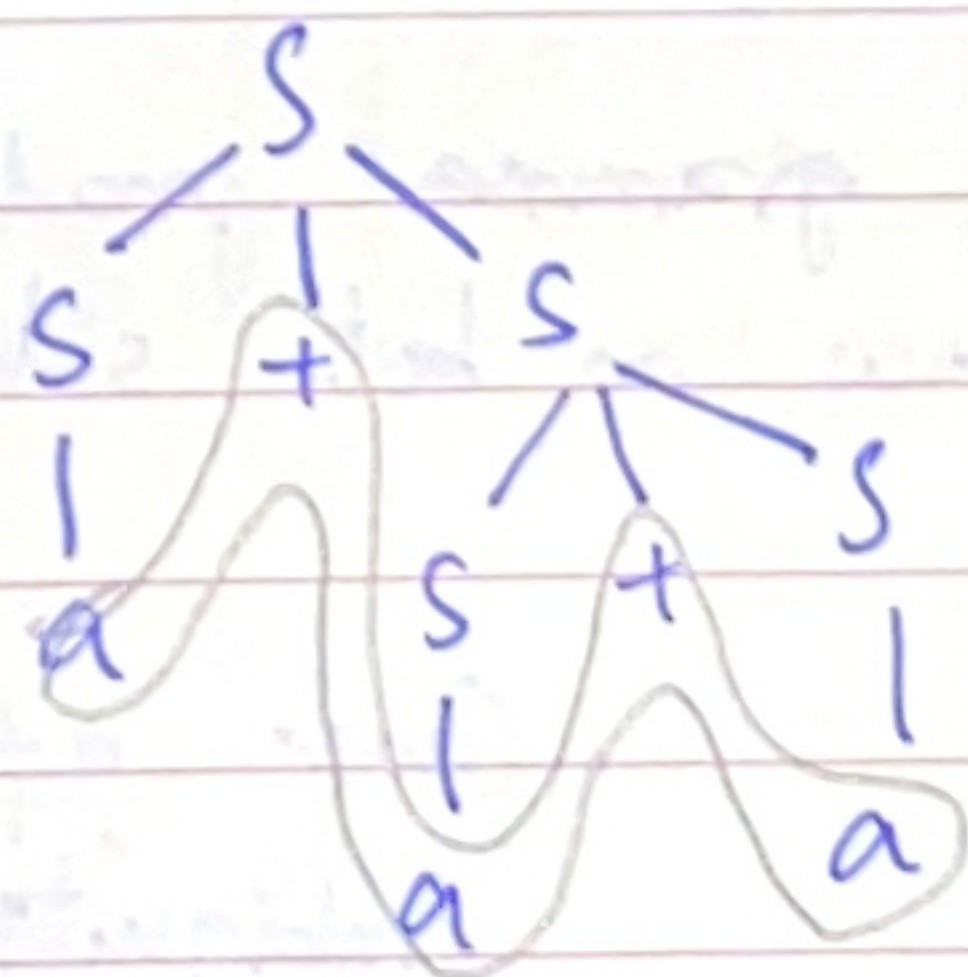
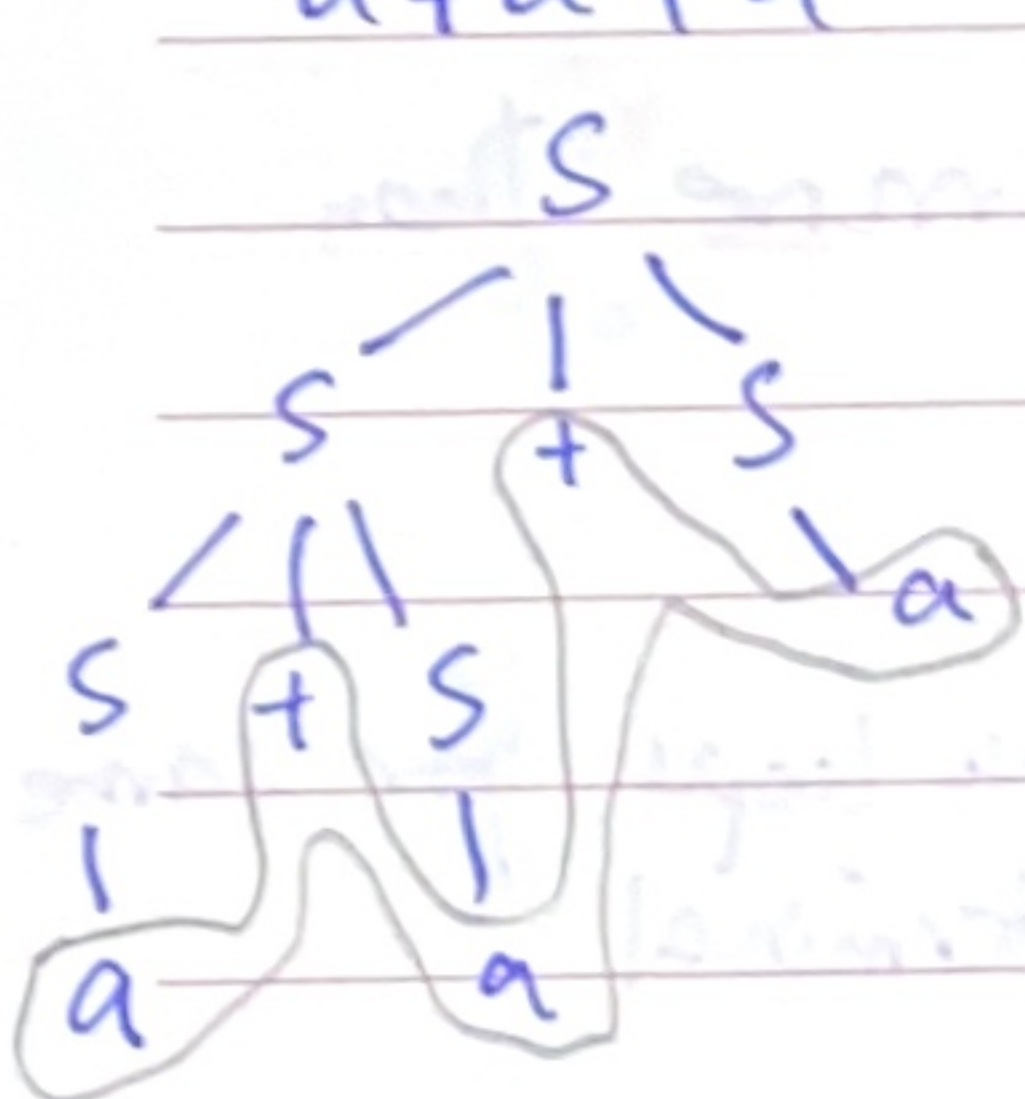


Removing : $S \rightarrow \Lambda / A$
 Ambiguity $A \rightarrow aAb / a\bar{b}A / ab$

④ $S \rightarrow S+S / a$
 $a+a+a$

$$\Sigma = \{a, +\}$$

replace 's' with a terminator



- Remove Ambiguity: $S \rightarrow a + S / a$

- Cause of: 'S' on both sides of '+'. Ambiguity